

A Metric with Positive Sectional Curvature on $S^3 \times S^3$

ABSTRACT

We construct a smooth Riemannian metric with strictly positive sectional curvature on $S^3 \times S^3$. Following the third-order deformation method of Brendle and Hung [BH26], we perturb a nonnegatively curved metric obtained by two diagonal Cheeger deformations. Minimization in directions normal to the zero-curvature locus in the bundle of tangent two-planes gives a nonnegative second-order coefficient whose zeros are the tangent planes of one flat torus. The reduced cubic coefficient has positive average on this torus, and a conformal correction gives strict positivity everywhere. The curvature identities used in the proof are verified by exact symbolic calculations supplied with the paper. Since $S^3 \times S^3$ is even-dimensional and has Euler characteristic zero, the result contradicts the positive-curvature case of Hopf's Euler characteristic conjecture.

1 Introduction

Products of spheres are natural test cases for the relation between sectional curvature and topology. Their product metrics have nonnegative curvature, but every mixed two-plane is flat. Two conjectures of Hopf provide distinct perspectives: the product conjecture excludes positive sectional curvature on nontrivial compact products, whereas the Euler-characteristic conjecture predicts positive Euler characteristic for closed even-dimensional positively curved manifolds; see [Zil07, Section 1] for their classical formulations. The case $S^3 \times S^3$ connects these two questions.

The variational approach has a long history. Berger's first-order rigidity for compact product metrics is recalled in [BDS72, Section 1.1]. Bourguignon, Deschamps, and Sentenac developed a higher-order theory for product metrics without infinitesimal isometries [BDS72, Section 7]; that additional hypothesis is essential and is not imposed here. Strake showed how totally geodesic immersions transmit first curvature variations and yield obstructions from flat tori [Str87, Lemma 4.1 and Proposition 4.3]. Gauss–Bonnet forces a nonnegative first variation on all tangent planes of a totally geodesic flat torus to vanish identically.

In this paper we prove the following theorem.

Theorem 1.1. *The smooth manifold $S^3 \times S^3$ admits a smooth Riemannian metric with strictly positive sectional curvature.*

Throughout the paper, $M = S^3 \times S^3$. Since $\dim M = 6$ and $\chi(M) = 0$, Theorem 1.1 provides a counterexample to the positive-curvature case of Hopf's Euler characteristic conjecture, which asserts $\chi(M) > 0$ for closed even-dimensional M with $\sec > 0$. This is the strict inequality, not the nonnegative-curvature prediction $\chi(M) \geq 0$.

For $S^3 \times S^3$, Spatzier and Strake observed that the normal homogeneous metric on $SU(2)^3/\Delta SU(2)$ has rank one in the sense of parallel Jacobi fields [SS90, Remark 2.4]; it nevertheless retains flat tori. Their first-order rigidity theory provides further context [SS90, Section 5]. Our background is obtained by Cheeger deformations. We use M\"uter's analysis of their curvature and zero planes, as presented by Ziller [Zil07, Section 2] and [Zil09].

The third-order deformation strategy used in this proof is due to Brendle and Hung [BH26]. In their construction on $S^2 \times S^2$, they first replace the product by a Cheeger–M\"uter metric and then deform it through third order. They analyze the curvature after minimizing over nearby

planes, arrange that its second coefficient is nonnegative and vanishes only on a distinguished torus, and use a nonzero cubic mean and a conformal Poisson correction to remove the remaining degeneracy [BH26, Sections 2 and 5–7]. We adopt this strategy, including its treatment of the second- and third-order curvature coefficients and its use of the full manifold of tangent two-planes near exceptional points.

The earlier $S^2 \times S^3$ construction [GFL26, Sections 6–8] uses a global fiber-rotating tensor and an anisotropic nearby-plane estimate to obtain positivity at second order. Its introduction gives further background on sphere products and weaker curvature conditions. Here we work directly on $S^3 \times S^3$, not by lifting that five-dimensional metric. Neither of the two preceding existence theorems is a hypothesis of our construction.

Two successive diagonal Cheeger deformations give a background on $S^3 \times S^3$ whose zero-curvature locus has two six-dimensional components with different projection behavior. We implement the Brendle–Hung strategy on these two components. One tensor makes the reduced second-order coefficient strictly positive on the second component without changing it on the first; a height-tensor perturbation gives a quadratic margin on the first family away from one torus. Further tensors produce a positive average of the reduced cubic coefficient on that torus. A global mixed correction preserves the quadratic margin. We give geometric and algebraic versions.

The construction uses explicit rational tensor formulas, followed by a mixed correction and auxiliary scale choices defined by uniform bounds on the compact zero locus. These choices determine smooth symmetric tensors H, K and a smooth function $\tilde{f}$ such that

$$g_\tau = g_0 + \tau H + \tau^2 K + \tau^3 \tilde{f} g_0, \quad 0 < \tau < \tau_0, \quad (1.1)$$

has positive sectional curvature. On $\mathcal{N} = \text{Gr}_2(TM)$, the background zero set $\mathcal{Z} = \mathcal{Z}_I \sqcup \mathcal{Z}_{II}$ has a positive sectional-curvature Hessian in its eight normal directions. Following Brendle–Hung’s notation, we write the normal minimum for $g_0 + \tau H + \tau^2 K$ as

$$\tau^2 V^{(2)}(z) + \tau^3 V^{(3)}(z) + O(\tau^4), \quad z \in \mathcal{Z}, \quad (1.2)$$

with a remainder uniform on $\mathcal{Z}$. Section 3 explains the normalization as sectional curvature. We arrange $V^{(2)} \geq 0$, with zero set $\hat{T}_0$, the tangent-plane lift of one flat torus T_0 . The average of the pullback of $V^{(3)}$ to T_0 is $512\varepsilon^3/375 > 0$, where ε is a fixed small auxiliary coefficient. Solving a scalar Poisson equation on T_0 makes the cubic coefficient $\hat{V}^{(3)}$ equal to this positive constant on $\hat{T}_0$.

Two points require care. Fixed-base plane variations miss one normal direction on the type-II component and two on the type-I component above the diagonal and antidiagonal, $p = \pm q$. We use the full base-and-plane bundle and a smooth minimizing displacement extending across these loci. The mixed correction must also cancel first transverse derivatives along $\hat{T}_0$. Tangent–normal tensor components and an antiperiodic integral provide a smooth correction, including at $p = \pm q$. Its parity preserves the cubic mean.

All auxiliary constants are fixed before τ tends to zero. The estimates therefore give one $\tau_0 > 0$ valid for every base point and every two-plane.

Section 2 identifies the background and its zero-curvature planes. Section 3 states the normal-minimum formulas and the flat-torus linearization. Sections 4 and 5 construct the tensors and establish the quadratic margin. Section 6 computes the cubic mean, and Section 7 completes the construction. Appendix A gives the exact calculation conventions, an algebraic mixed correction, and the finite verification procedure. These exact computations and the geometric argument constitute the proof presented here. As in [BH26], the paper states the identities and their role in the argument, and supplies code for the longer algebraic verifications. Floating-point cross-checks are not used to establish them.

2 The background metric and its zero-curvature planes

We represent S^3 by the unit quaternions and write

$$p = p_0 + p_1i + p_2j + p_3k, \quad q = q_0 + q_1i + q_2j + q_3k.$$

For an imaginary quaternion a , put $E_a = (pa, 0)$ and $F_a = (0, qa)$. The combined left Maurer–Cartan coframe is denoted by $(\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3)$. The corresponding right coframes are obtained by applying the adjoint actions:

$$\hat{\alpha} = \text{Ad}_p \alpha, \quad \hat{\beta} = \text{Ad}_q \beta.$$

Thus $[E_a, E_b] = E_{[a,b]}$, with $[a, b] = 2a \times b$, and similarly on the second factor. We use $\lambda \odot \mu = (\lambda \otimes \mu + \mu \otimes \lambda)/2$. Unless stated otherwise, matrices of covariant tensors are written in this combined left coframe.

Set $r = p^{-1}q = (r_0, v)$ and $R = \text{Ad}_r$. Define the background metric by its cometric:

$$g_0^{-1} = \frac{1}{30} \begin{pmatrix} 29I & 10I + 9R \\ 10I + 9R^T & 29I \end{pmatrix}. \quad (2.1)$$

Here I is the 3×3 identity; subscripts specify other sizes. The diagonal and antidiagonal are

$$\Delta_+ = \{(p, q) : q = p\}, \quad \Delta_- = \{(p, q) : q = -p\};$$

equivalently, $r = 1$ and $r = -1$, respectively.

Proposition 2.1. *The metric g_0 is smooth, positive definite, and nonnegatively curved. Its zero-curvature planes are exactly*

$$P_u = \text{span}\{E_u, F_u\}, \quad |u| = 1, \quad Ru = u \text{ or } Ru = -u. \quad (2.2)$$

Proof. On each acting copy of $SU(2)$, fix the bi-invariant metric Q for which i, j, k are orthonormal. Our Cheeger parameter convention is the quotient of $(M, g) \times (SU(2), t^{-1}Q)$ for $t > 0$. Begin with the bi-invariant product metric $3I_6$. A Cheeger deformation by simultaneous right multiplication, with parameter $1/3$, gives the metric

$$g_1 = \begin{pmatrix} 2I & -I \\ -I & 2I \end{pmatrix}. \quad (2.3)$$

A second deformation, now by simultaneous left multiplication and with parameter $3/10$, adds $\frac{3}{10} \begin{pmatrix} I & R \\ R^T & I \end{pmatrix}$ to the cometric. This gives (2.1). These assertions follow either from the defining submersion or from the metric-operator formula in [Zil09, Proposition 1.1]. Nonnegative curvature follows from the submersion formula. The singular values of $10I + 9R$ are at most 19, so the cometric eigenvalues lie in $[1/3, 8/5]$; in particular $g_0 \geq (5/8)I_6$.

A zero-curvature plane for the product metric is mixed. In the reparametrized Cheeger formula [Zil09, Proposition 1.3], the first group-bracket term forces its two axes to be parallel. The remaining planes are precisely P_u . They are flat for g_1 : the corresponding abelian two-dimensional subgroups, and their left translates, are totally geodesic by the Koszul formula. For the second deformation, the two Lie algebra vectors in the group-bracket term are $2\text{Ad}_p u - \text{Ad}_q u$ and $2\text{Ad}_q u - \text{Ad}_p u$. After conjugation by p^{-1} , their bracket is a nonzero multiple of

$$(2u - Ru) \times (-u + 2Ru) = 3u \times Ru.$$

Every summand of the reparametrized curvature formula is nonnegative, so a zero-curvature plane must satisfy $Ru = \pm u$. At these planes the reparametrization preserves P_u : it rescales an orbit direction in that plane and fixes its horizontal direction. The flat tori identified below prove that each plane satisfying this condition is indeed flat. $\square$

Let $\text{Gr}_2(\mathbb{R}^4)$ denote the unoriented Grassmannian. For $L \in \text{Gr}_2(\mathbb{R}^4)$, put $C_L = L \cap S^3$. Define two subsets of $\mathcal{N} = \text{Gr}_2(TM)$ by

$$\mathcal{Z}_I = \{T_{(p,q)}(C_L \times C_L) : p, q \in C_L\}, \quad (2.4)$$

$$\mathcal{Z}_{II} = \{T_{(p,q)}(C_L \times C_{L^\perp}) : p \in C_L, q \in C_{L^\perp}\}. \quad (2.5)$$

Each set includes all $L \in \text{Gr}_2(\mathbb{R}^4)$. We call the corresponding tori type I and type II, respectively, and set

$$\mathcal{Z}_I^{\text{reg}} = \{((p, q), P_u) \in \mathcal{Z}_I : p \neq \pm q\}.$$

Proposition 2.2. *The zero set of the sectional-curvature function on $\mathcal{N}$ is $\mathcal{Z} = \mathcal{Z}_I \sqcup \mathcal{Z}_{II}$. Its two components are disjoint compact smooth six-dimensional submanifolds. Each torus in (2.4)–(2.5) is flat and totally geodesic. The sectional-curvature Hessian is positive definite on the eight-dimensional normal bundle of $\mathcal{Z}$. Fix a smooth auxiliary Riemannian metric on $\mathcal{N}$, and let dist denote its induced distance. There is $b_0 > 0$, depending on this choice, such that*

$$\sec_{g_0}(\sigma) \geq b_0 \text{dist}(\sigma, \mathcal{Z})^2 \quad \text{for all } \sigma \in \text{Gr}_2(TM). \quad (2.6)$$

Proof. If $Ru = u$, then $r \in \text{span}\{1, u\}$, and both p, q lie in the plane $L = \text{span}\{p, pu\}$. If $Ru = -u$, then $r_0 = 0$ and $v \perp u$; hence $q, qu \in L^\perp$. This identifies (2.2) with the two incidence descriptions. The point and lifted plane recover L smoothly: the projection of P_u to the first tangent factor has constant rank one, and adjoining the vector p recovers L . Thus these compact incidence bundles embed in $\text{Gr}_2(TM)$. Each has base dimension four and fiber dimension two. The signs in (2.2) are mutually exclusive, so the components are disjoint.

The metric is invariant under the diagonal $SO(4)$ -action and under either factor's antipodal map. The orthogonal map equal to $+I$ on L and $-I$ on $L^\perp$ belongs to $SO(4)$; its diagonal action fixes $C_L \times C_L$. Composing its action on the second factor with the antipodal map fixes $C_L \times C_{L^\perp}$. These fixed sets are totally geodesic. In the frames E_u, F_u , their constant metric matrices are

$$G_I = \frac{1}{16} \begin{pmatrix} 29 & -19 \\ -19 & 29 \end{pmatrix}, \quad G_{II} = \frac{1}{28} \begin{pmatrix} 29 & -1 \\ -1 & 29 \end{pmatrix}. \quad (2.7)$$

Their determinants are $D_I = 15/8$ and $D_{II} = 15/14$. Constancy proves flatness. Reversing one great-circle coordinate changes the sign of the off-diagonal entry but does not change the determinant.

For the Hessian calculation, all terms must be compared in the same variables. Let C_t be the positive metric operator defined by $g_t(V, W) = g_{\text{old}}(C_t V, W)$. For a Cheeger deformation, the map

$$\Psi_t : \text{Gr}_2(TM) \longrightarrow \text{Gr}_2(TM), \quad (x, \sigma) \longmapsto (x, C_t^{-1} \sigma)$$

is a smooth diffeomorphism covering the identity of M . Pull back the second curvature formula by the composition of the two such maps, and substitute the first curvature formula into its old-curvature term. Locally choosing a smooth basis of each plane gives a sum of nonnegative terms, up to positive smooth Gram factors. At a zero of a squared norm $\|B\|^2$, its Hessian is $2(dB)^* dB$; positive factors do not change its kernel. We check each new bracket differential on the common kernel of the preceding terms.

For the product metric, a mixed plane has a basis $(a, 0), (0, b)$. Its graph variations have four components that move these two axes within their own factors, and four that mix the factors. The latter give the positive quadratic terms $\|a \times \delta Y_1\|^2 + \|\delta X_2 \times b\|^2$, up to positive constants. Thus the product Hessian has rank four, with kernel the tangent space to the ten-dimensional mixed-plane locus. On that locus the first group bracket is a nonzero

multiple of $a \times b$. At parallel axes its differential has rank two in the relative-axis directions. Its kernel consists of the common-axis variations and all base-point directions, giving the eight-dimensional zero locus of g_1 , parametrized by $(p, q, [u])$ and the plane P_u . The first reparametrization fixes each such plane, so these are also coordinates for the remaining common kernel.

On this kernel the second group bracket has the same differential rank as $\Phi(r, u) = u \times \text{Ad}_r u$. At a type-I zero $r = (a, bu)$, $a^2 + b^2 = 1$, varying v by $w \perp u$ gives

$$D(u \times Ru)[w] = 2(aI + bJ_u)w, \quad J_u w = u \times w. \quad (2.8)$$

This map is invertible on $u^\perp$, including when $r = \pm 1$. At a type-II zero, variations of r_0 and $v \cdot u$ give the independent images $2v$ and $2u \times v$. The second bracket thus has rank two on the remaining kernel everywhere. The successive common kernels have dimensions 10, 8, 6, respectively. The known six-dimensional zero locus is contained in the kernel of the full curvature Hessian, while these nonnegative terms already give rank eight. Hence the full kernel is exactly $T\mathcal{Z}$, and the Hessian is positive definite on a normal complement. Transport by the two diffeomorphisms above gives the same conclusion in the original variables.

Taylor's theorem gives a uniform quadratic lower bound on a tube about the compact zero set. On the compact complement, curvature has a positive minimum. Decreasing the constant gives (2.6) on the whole Grassmann bundle. $\square$

The intermediate metric (2.3) also has the quotient description in the introduction. The quotient of the product bi-invariant metric on $SU(2)^3$ has, in the coordinates $[g_1, g_2, g_3] \mapsto (g_1 g_3^{-1}, g_2 g_3^{-1})$, matrix $\frac{1}{3} \begin{pmatrix} 2I & -I \\ -I & 2I \end{pmatrix}$. Thus $g_1/3$ is precisely this quotient metric.

3 Normal minimization of sectional curvature

We use the minimum-curvature framework of Brendle and Hung [BH26, Lemma 2.1 and Proposition 2.9]. We include the versions of their expansion and change-of-coordinate arguments needed here. The formulation keeps the base point and the plane in a single compact manifold. This is needed both over $\Delta_+ \cup \Delta_-$ and on the type-II component. The ranks of the curvature Hessian restricted to the eight fixed-base plane directions are

zero-plane locus	fixed-base rank	additional base directions
$\mathcal{Z}_I^{\text{reg}}$	8	0
$\mathcal{Z}_I$ over $(\Delta_+ \cup \Delta_-)$	6	2
$\mathcal{Z}_{II}$	7	1

Indeed, after the six directions supplied by the first deformation, the remaining fixed-base variations are $u \mapsto u + w$, $w \perp u$. For $\Phi(u) = u \times Ru$, their differential is

$$D\Phi(w) = \begin{cases} J_u(R - I)w, & Ru = u, \\ J_u(R + I)w, & Ru = -u. \end{cases}$$

The first map has rank two off $\Delta_+ \cup \Delta_-$ and rank zero on it; the second has rank one, since R is a half-turn. The base variations used in Proposition 2.2 supply the missing directions. Appendix A.4 gives explicit transverse slices at the exceptional representatives.

For a Riemannian metric g with Levi-Civita connection ∇ , we use the curvature endomorphism

$$R^g(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z. \quad (3.1)$$

Thus the numerator of sectional curvature is $g(R^g(X, Y)Y, X)$. This is the quantity denoted by $R_g(X, Y, X, Y)$ in the covariant curvature convention of [BH26, Appendix A]; more

generally, their convention is $R_g(X, Y, Z, W) = g(R^g(X, Y)W, Z)$. All series below use Taylor coefficients, so the coefficient of order j is the j -th derivative divided by $j!$.

Lemma 3.1 (Normal minimum). *Let N be a compact manifold, and let $Z \subset N$ be a compact embedded submanifold. Suppose a smooth family of functions has expansion*

$$F_\tau = F_0 + \tau F_1 + \tau^2 F_2 + \tau^3 F_3 + O(\tau^4),$$

where $F_0 \geq 0$, $F_0^{-1}(0) = Z$, the Hessian of F_0 is positive definite on the normal bundle of Z , and $F_1|_Z = 0$. Fix tubular coordinates (z, ν) near Z . For sufficiently small $|\tau|$, minimization in ν gives a smooth function

$$F_{\min}(\tau, z) = \tau^2 q(z) + \tau^3 r(z) + O(\tau^4), \quad (3.2)$$

uniformly on Z . If $A_z = D_\nu^2 F_0(z, 0)$ and $b_z = D_\nu F_1(z, 0)$, then the first minimizing displacement and second coefficient are

$$\eta_1 = -A_z^{-1} b_z, \quad q = F_2|_Z - \frac{1}{2} b_z^T A_z^{-1} b_z. \quad (3.3)$$

The cubic coefficient is

$$r = F_3 + D_\nu F_2[\eta_1] + \frac{1}{2} D_\nu^2 F_1[\eta_1, \eta_1] + \frac{1}{6} D_\nu^3 F_0[\eta_1, \eta_1, \eta_1], \quad (3.4)$$

with all quantities evaluated at $(z, 0)$. If $q \geq 0$ on Z and $r > 0$ on $q^{-1}(0)$, then $F_\tau > 0$ on N for every sufficiently small $\tau > 0$.

Proof. Uniform positivity of the normal Hessian and the implicit function theorem give a smooth critical section $\nu_\tau(z)$, with $\nu_\tau = \tau \eta_1 + O(\tau^2)$. On a sufficiently small fixed tube the normal Hessian remains positive for small τ , so this critical section is the normal minimum. Differentiating the critical equation gives $A_z \eta_1 + b_z = 0$. Substitution gives (3.3). In the cubic expansion, every term involving the second displacement has the factor $A_z \eta_1 + b_z$; these terms cancel, leaving (3.4). Compactness makes the remainders uniform.

If $q^{-1}(0)$ is nonempty, choose a neighborhood of it in Z on which r has a positive lower bound. There the cubic term dominates the fourth-order remainder. On the compact remainder of Z , the positive minimum of q dominates the higher terms. The same conclusion holds immediately if $q^{-1}(0)$ is empty. Positivity of each normal minimum then gives positivity on the tube. On the compact complement, the positive minimum of F_0 persists. $\square$

Lemma 3.2 (Change of normal section). *Under the hypotheses of Lemma 3.1, the second reduced coefficient is independent of the tubular choice. Two first minimizing displacements differ by a tangent vector to Z . At a point where $q = 0$ and $dq = 0$ along Z , they give the same cubic coefficient. Changing the second displacement also does not change that coefficient.*

Proof. The Hessian of F_0 has kernel TZ , and dF_1 annihilates TZ . Thus (3.3) defines a unique class of first displacements modulo TZ , and its value q is independent of the chosen complement. If a second choice adds $t \in T_z Z$, its footpoint is $z + \tau t + O(\tau^2)$. The reduced value is therefore

$$\tau^2 q(z) + \tau^3 (r(z) + dq_z[t]) + O(\tau^4).$$

This proves the asserted invariance. The cancellation of the second displacement is the stationarity cancellation in (3.4).

To compare normalizations along a fixed first minimizing curve, write

$$N_\tau = \tau^2 n_2 + \tau^3 n_3 + O(\tau^4), \quad D_\tau = D_0 + \tau d_1 + O(\tau^2), \quad D_0 > 0.$$

Then

$$[\tau^3] \frac{N_\tau}{D_\tau} = \frac{n_3}{D_0} - \frac{n_2 d_1}{D_0^2}.$$

Thus, when $n_2 = 0$, two denominators with the same positive zeroth-order value give the same cubic coefficient. The factor $1/D_0$ remains. The assumption $q = 0$ is precisely the vanishing of the second numerator coefficient along the minimizing curve. $\square$

Apply this discussion with $N = \mathcal{N} = \text{Gr}_2(TM)$ and $F_\tau(\sigma) = \sec_{g_0+\tau h+\tau^2 k}(\sigma)$. For a smooth symmetric tensor h , write $F_{1,h}(\sigma) = [\tau] \sec_{g_0+\tau h}(\sigma)$ on $\mathcal{N}$, and define

$$\mathcal{L}h = [\tau] \sec_{g_0+\tau h} \big|_{\mathcal{Z}}. \quad (3.5)$$

When $h \in \ker \mathcal{L}$, write $\mathcal{Q}(h)$ for the second coefficient after normal minimization with $k = 0$. This is quadratic in h ; we use the polarization

$$\mathcal{Q}(h_1, h_2) = \frac{1}{2}(\mathcal{Q}(h_1 + h_2) - \mathcal{Q}(h_1) - \mathcal{Q}(h_2)), \quad h_1, h_2 \in \ker \mathcal{L}.$$

For the path with both tensors, the normal minimum is

$$\tau^2(\mathcal{Q}(h) + \mathcal{L}k) + \tau^3 \mathcal{R}(h, k) + O(\tau^4). \quad (3.6)$$

We fix one system of tubular coordinates when $\mathcal{R}(h, k)$ is used away from a critical zero of the second coefficient.

The calligraphic operators $\mathcal{L}, \mathcal{Q}, \mathcal{R}$ take values in scalar functions on $\mathcal{Z}$. They are distinct from the tensor-valued operators L, Q, C in [BH26, Section 4], which act before projection to a two-plane and normal minimization. In particular, $\mathcal{Q}$ includes the Schur term in (3.3). Our symbols $V^{(2)}$ and $V^{(3)}$ denote the resulting coefficients for the final pair H, K .

There is also a normalization difference. The global curvature function in [BH26, Proposition 5.3] uses the fixed background Gram determinant. Here F_τ is sectional curvature and uses the perturbed Gram determinant. For a fixed background and metric path, the normal minima of the two normalized functions have the same reduced second coefficient on $\mathcal{Z}$ when $\mathcal{L}h = 0$, and the same reduced third coefficient where that second coefficient vanishes. Independence of the normal coordinates also requires its differential to vanish, as in Lemma 3.2. For an unnormalized curvature numerator one must additionally divide by the background determinant; the factors D_I and D_{II} below make this conversion explicit.

Let D_h denote the first variation of the Levi-Civita connection. In terms of the background connection ∇ ,

$$2g_0(D_h(X, Y), W) = (\nabla_X h)(Y, W) + (\nabla_Y h)(X, W) - (\nabla_W h)(X, Y). \quad (3.7)$$

At a point of $\mathcal{Z}_I^{\text{reg}}$, after diagonal $SO(4)$ -normalization, choose $X = E_i$, $Y = F_i$ and the transverse frame $N = (E_j, E_k, F_j, F_k)$. Let A be the Hessian of the unnormalized background curvature in the eight coefficients of $X + a \cdot N, Y + b \cdot N$. Let r_h be the gradient in those coefficients of the first metric variation of this same *unnormalized curvature numerator*, evaluated at $a = b = 0$. For $h_1, h_2 \in \ker \mathcal{L}$,

$$\begin{aligned} D_I \mathcal{Q}(h_1, h_2) &= g_0(D_{h_1}(X, Y), D_{h_2}(X, Y)) \\ &\quad - \frac{1}{2} g_0(D_{h_1}(X, X), D_{h_2}(Y, Y)) \\ &\quad - \frac{1}{2} g_0(D_{h_2}(X, X), D_{h_1}(Y, Y)) - \frac{1}{2} r_{h_1}^T A^{-1} r_{h_2}. \end{aligned} \quad (3.8)$$

This is the metric second-variation formula followed by (3.3). In particular it includes the loss from moving the plane. At degenerate fixed-base points the definition (3.6), with a full base-and-plane normal complement, replaces this particular coordinate formula.

On a background flat torus, let s, t be great-circle parameters of unit speed with respect to the round metrics on the two factors, and put $X = \partial_s$, $Y = \partial_t$. These vectors need not have unit g_0 -length. Total geodesicity and the linearized Gauss equation give

$$\mathcal{L}k = \frac{1}{D} \left(\partial_s \partial_t k(X, Y) - \frac{1}{2} \partial_s^2 k(Y, Y) - \frac{1}{2} \partial_t^2 k(X, X) \right), \quad (3.9)$$

where

$$D = g_0(X, X)g_0(Y, Y) - g_0(X, Y)^2 = D_I \text{ or } D_{II}.$$

Remark 3.3. If $g_0 + \tau h + O(\tau^2)$ has positive sectional curvature for all small positive τ , then $\mathcal{L}h \geq 0$ on each flat torus. The first ambient variation there equals the intrinsic Gaussian-curvature variation, because the initial second fundamental form vanishes. Gauss–Bonnet gives $\int_T \mathcal{L}h \, dA_{g_0} = 0$, so $\mathcal{L}h = 0$ everywhere on T . Thus the vanishing first-variation condition used in (3.6) is forced, not an incidental feature of our ansatz.

4 The perturbation tensors

We first give the algebraic tensors used in the construction. Their formulas involve only quaternion coordinates and the global left and right coframes. The remaining mixed correction will be constructed geometrically in Section 5.1, without choosing an axis at a singular point. For $v \in \mathbb{R}^3$, set $P(v) = |v|^2 I - vv^T$. Define

$$H_0 = r_0 \begin{pmatrix} 0 & P(v) \\ P(v) & 0 \end{pmatrix}, \quad (4.1)$$

$$O = p_0 \begin{pmatrix} I & -\frac{2}{7}I \\ -\frac{2}{7}I & 0 \end{pmatrix} + q_1 \begin{pmatrix} 0 & -\frac{2}{7}I \\ -\frac{2}{7}I & I \end{pmatrix}, \quad (4.2)$$

$$B = \frac{1}{2}(\alpha_1 + \hat{\alpha}_1) \odot \left(\alpha_2 + \frac{3}{13}\beta_2 \right) - \frac{1}{2}(\beta_1 + \hat{\beta}_1) \odot \left(\frac{3}{13}\alpha_2 + \beta_2 \right), \quad (4.3)$$

$$C = \alpha_1^2. \quad (4.4)$$

The tensor H_0 makes the reduced second-order coefficient strictly positive on $\mathcal{Z}_{II}$ without affecting it on $\mathcal{Z}_I$. The tensor O , odd under the simultaneous antipodal map, will supply the principal quadratic margin. The smaller combination $B + C$ supplies a positive average of the reduced cubic coefficient on the residual torus.

The second-order corrections are

$$K_0 = -\frac{1}{5}p_0q_1 \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} - \frac{787}{5880}p_1q_0I_6, \quad (4.5)$$

$$K_2 = -\frac{96}{845}(2r_0^2 - 1)g_0, \quad (4.6)$$

$$K_3 = \frac{8}{5}(Bg_0^{-1}C + Cg_0^{-1}B). \quad (4.7)$$

The matrix products in (4.7) mean contraction through the background cometric, so their sum is a globally defined symmetric tensor.

Lemma 4.1. *The smooth tensors H_0, O, B, C satisfy $\mathcal{L}H_0 = \mathcal{L}O = \mathcal{L}B = \mathcal{L}C = 0$ on $\mathcal{Z}$. The two height summands O_p, O_q in (4.2) separately satisfy $\mathcal{L}O_p = \mathcal{L}O_q = 0$.*

Proof. Smoothness is immediate from the global formulas. On a type-I torus the left great-circle axes and right great-circle axes are each constant. The same is true on a type-II torus, with opposite right axes on its two factors. Thus the pullbacks of B and C have constant

tangential components in the round-unit-speed circle coordinates, and (3.9) vanishes. For O , its tangent entries in the frame E_u, F_u are

$$O_{ss} = p_0(s), \quad O_{tt} = q_1(t), \quad O_{st} = -\frac{2}{7}(p_0(s) + q_1(t)).$$

Each derivative in (3.9) is again zero, also for either summand separately. On a type-I torus, v is parallel to u , so $P(v)u = 0$; on a type-II torus, $r_0 = 0$. Hence its restriction vanishes on both families, as required. $\square$

Proposition 4.2. *On $\mathcal{Z}_I$,*

$$\mathcal{Q}(H_0, h) = 0 \quad \text{whenever } \mathcal{L}h = 0. \quad (4.8)$$

On the second family,

$$\mathcal{Q}(H_0) = \frac{282877}{2278125} > \frac{1}{10}. \quad (4.9)$$

Proof. The tensor H_0 is diagonal $SO(4)$ -invariant. Every type-I torus therefore remains totally geodesic for $g_0 + \tau H_0$, with unchanged flat induced metric. For the commuting round-unit-speed coordinate fields X, Y on such a torus, the ambient derivatives $\nabla_X^{g_0 + \tau H_0} X$, $\nabla_X^{g_0 + \tau H_0} Y$, and $\nabla_Y^{g_0 + \tau H_0} Y$ vanish for every small τ . Hence

$$D_{H_0}(X, X) = D_{H_0}(X, Y) = D_{H_0}(Y, Y) = 0 \quad \text{as ambient vectors.}$$

Flatness and total geodesicity also give zero curvature differential in the plane directions. At points of $\mathcal{Z}_I^{\text{reg}}$, these directions together with $T\mathcal{Z}_I$ span the ambient tangent space. Since $F_{1, H_0}|_{\mathcal{Z}} = 0$, it follows that $dF_{1, H_0} = 0$ there, and continuity extends this to all of $\mathcal{Z}_I$. Formula (3.8) proves (4.8).

The diagonal $SO(4)$ -action is transitive on the second zero family and preserves H_0 , so it suffices to use $p = 1, q = j, u = i$. At this point $H_0 = 0$ and

$$dH_0 = (\alpha_2 - \beta_2) \otimes \begin{pmatrix} 0 & P(j) \\ P(j) & 0 \end{pmatrix}.$$

Consequently $D_{H_0}(E_i, E_i) = D_{H_0}(F_i, F_i) = 0$, while $D_{H_0}(E_i, F_i) = -\frac{1}{2} \text{grad}_{g_0} r_0$. Since $|dr_0|_{g_0}^2 = 2/3$, the fixed quadratic numerator is $1/6$. The full normal Schur subtraction is $5107/151875$, as computed in Appendix A; the calculation includes the base direction changing r_0 . Dividing by D_{II} gives

$$\mathcal{Q}(H_0) = \frac{1/6 - 5107/151875}{15/14} = \frac{282877}{2278125}.$$

Transitivity can also be seen from the adapted orthonormal frames (p, pu, q, qu) : all have the orientation of $(1, i, j, -k)$, and hence lie in a single $SO(4)$ -orbit. $\square$

5 The second-order curvature coefficient

We choose the first two perturbation tensors so that the second-order coefficient is nonnegative on $\mathcal{Z}$, with zero set the tangent planes to one flat torus. The principal estimate is the following calculation for the height tensor O ; the mixed corrections will be chosen to preserve it.

Let e_0, e_1, e_2, e_3 be the standard basis of $\mathbb{R}^4$, and put

$$L_0 = \text{span}\{e_0, e_1\}, \quad T_0 = C_{L_0} \times C_{L_0}.$$

Distinguish this torus from its tangent-plane lift by writing

$$\iota_0 : T_0 \longrightarrow \mathcal{Z}_I, \quad (p, q) \longmapsto ((p, q), T_{(p, q)}T_0), \quad \widehat{T}_0 = \iota_0(T_0).$$

For $L \in \text{Gr}_2(\mathbb{R}^4)$, let $e_\perp, f_\perp$ be the projections of e_0, e_1 onto $L^\perp$, and set

$$\rho(L) = |e_\perp|^2 + |f_\perp|^2. \quad (5.1)$$

This is a smooth function on the Grassmannian, vanishing precisely at L_0 . Its pullback to $\mathcal{Z}_I$ is comparable to the squared transverse distance to $\widehat{T}_0$ in $\mathcal{Z}_I$.

Proposition 5.1 (Quadratic lower bound for the odd height tensor). *On a type-I torus, choose one oriented round-unit-speed parametrization $c : \mathbb{R}/2\pi\mathbb{Z} \rightarrow C_L$, write $p = c(s)$, $q = c(t)$, and set $\theta = t - s$, $x = \sin^2 \theta$. Define the scalar polynomials*

$$\begin{aligned} d(x) &= 31752000(25 + 3x), \\ P(x) &= 12632468 + 1767741x + 5670x^2, \\ V(x) &= 3941932 + 686019x, \\ W(x) &= 3521524 + 636555x. \end{aligned}$$

Let J be the positive quarter-turn on $L^\perp$, using the orientation induced by the chosen orientation of L and the ambient orientation of $\mathbb{R}^4$. Then

$$\begin{aligned} \mathcal{Q}(O) + \mathcal{L}K_0 &= \frac{P(x)}{2d(x)}\rho + \frac{V(x)\cos\theta}{d(x)}\langle e_\perp, f_\perp \rangle \\ &\quad - \frac{W(x)\sin\theta}{d(x)}\langle Je_\perp, f_\perp \rangle. \end{aligned} \quad (5.2)$$

In particular,

$$\mathcal{Q}(O) + \mathcal{L}K_0 \geq \frac{1}{250}\rho \quad \text{on } \mathcal{Z}_I. \quad (5.3)$$

Proof. Use the diagonal $SO(4)$ -action to put $p = 1$, $q = \cos\theta + i\sin\theta$. The two fixed height vectors become arbitrary orthonormal vectors $e, f \in \mathbb{R}^4$. Formula (3.8) is quadratic in these height vectors. The two self-coefficient matrices and the corrected cross-coefficient matrix are computed in Appendix A. They give (5.2); the scalar multiple of $\langle e, f \rangle$ vanishes because $e \perp f$. This calculation is an exact identity over $\mathbb{Q}(z)$, with $\cos\theta = (1 - z^2)/(1 + z^2)$ and $\sin\theta = 2z/(1 + z^2)$. Smoothness of the reduced coefficient extends the identity to $\theta = 0, \pi$. Reversing the orientation of L changes both J and $\sin\theta$ in sign, so the formula is independent of that choice.

For $0 \leq x \leq 1$,

$$P(x) \geq 12632468, \quad \max\{V(x), W(x)\} \leq 4627951, \quad d(x) \leq 31752000 \cdot 28.$$

The vectors $e_\perp, Je_\perp$ are orthogonal and have equal length. Consequently the absolute value of the last two terms in (5.2) is at most $\max\{V, W\}|e_\perp||f_\perp|/d$. Since $2|e_\perp||f_\perp| \leq \rho$, the coefficient of ρ is bounded below by

$$\frac{12632468 - 4627951}{2 \cdot 31752000 \cdot 28} > \frac{1}{250}.$$

This proves (5.3). □

5.1 A global mixed correction

The interaction of O with B need not respect the quadratic margin near $\widehat{T}_0$. We construct a correction that cancels its first transverse jet. A tangent-normal tensor component is essential: at $p = \pm q$, motion of the supporting two-plane is not determined by motion of the base points alone.

The reflection

$$(p, q) \mapsto (ipi^{-1}, iqi^{-1}) \quad (5.4)$$

fixes $\widehat{T}_0$ and reverses its transverse directions in $\mathcal{Z}_I$; O, C, H_0 are even and B is odd.

Definition 5.2. A smooth symmetric tensor K_1 on M is an admissible mixed correction if it is odd under (5.4) and under $(p, q) \mapsto (-p, -q)$, and

$$S := 2\mathcal{Q}(O, B) + \mathcal{L}K_1 \quad \text{satisfies} \quad S|_{\widehat{T}_0} = 0, \quad dS|_{\widehat{T}_0} = 0. \quad (5.5)$$

The differential here is taken along $\mathcal{Z}_I$.

Proposition 5.3. *An admissible mixed correction exists. In particular, it can be chosen smoothly at every pair $p = \pm q$.*

Proof. Put $N = L_0^\perp$, and write

$$p_s = \cos s e_0 + \sin s e_1, \quad q_t = \cos t e_0 + \sin t e_1.$$

Parametrize nearby supporting planes by graphs $U : L_0 \rightarrow N$, with isometric inclusions

$$A_U = (I + U)(I + U^T U)^{-1/2}, \quad (p, q) = (A_U p_s, A_U q_t).$$

For the two height summands, set $\mathcal{S}_p = 2\mathcal{Q}(O_p, B)$ and $\mathcal{S}_q = 2\mathcal{Q}(O_q, B)$. These are smooth functions on the full zero family, by Section 3. Their reflection oddness gives zero on $\widehat{T}_0$. Write their first graph derivatives as

$$\mathcal{S}_p(s, t, U) = L_p(s, t)[U] + O(|U|^2), \quad \mathcal{S}_q(s, t, U) = L_q(s, t)[U] + O(|U|^2).$$

The remainder estimates hold with angular derivatives. The functional L_p is π -antiperiodic in s and π -periodic in t ; the parities of L_q are reversed. Indeed, O_p is odd only under the first-factor antipodal map, O_q only under the second, and B is even under both.

For a smooth π -antiperiodic function $f : \mathbb{R} \rightarrow \mathbb{R}$, define

$$\mathcal{I}f(s) = \int_0^\pi (\xi - \pi/2) f(s + \xi) d\xi.$$

Integration by parts and antiperiodicity give

$$(\mathcal{I}f)'(s) = - \int_0^\pi f(s + \xi) d\xi, \quad (\mathcal{I}f)''(s) = 2f(s).$$

Thus the smooth linear functionals

$$\begin{aligned} \kappa_p(s, t)[U] &= D_I \int_0^\pi (\xi - \pi/2) L_p(s + \xi, t)[U] d\xi, \\ \kappa_q(s, t)[U] &= D_I \int_0^\pi (\xi - \pi/2) L_q(s, t + \xi)[U] d\xi \end{aligned} \quad (5.6)$$

satisfy

$$\partial_s^2 \kappa_p = 2D_I L_p = \frac{15}{4} L_p, \quad \partial_t^2 \kappa_q = \frac{15}{4} L_q. \quad (5.7)$$

All parities are preserved. Smooth parameter dependence follows by differentiation under a fixed compact-interval integral. Write $\kappa_p[U] = \langle D_p, U \rangle_{\text{HS}}$ and $\kappa_q[U] = \langle D_q, U \rangle_{\text{HS}}$, where the linear maps $D_p, D_q : L_0 \rightarrow N$ depend smoothly on s, t .

In a tubular neighborhood of T_0 in M , use

$$p = \sqrt{1 - |x|^2} p_s + x, \quad q = \sqrt{1 - |y|^2} q_t + y, \quad x, y \in N.$$

Choose a smooth cutoff $\chi(|x|^2, |y|^2)$, equal to one near zero and supported inside this neighborhood. Define

$$\begin{aligned} k_p &= \chi[\langle D_p q_t, y \rangle dt^2 + dt \odot \langle D_p q'_t, dy \rangle], \\ k_q &= \chi[\langle D_q p_s, x \rangle ds^2 + ds \odot \langle D_q p'_s, dx \rangle], \\ K_1 &= k_p + k_q. \end{aligned} \tag{5.8}$$

Periodicity makes these genuine tensors on the neighborhood, and the cutoff gives a smooth extension by zero.

On the graph torus, with $X = \partial_s$ and $Y = \partial_t$,

$$y = U q_t + O(|U|^2), \quad dy(Y) = U q'_t + O(|U|^2), \quad dt(Y) = 1 + O(|U|^2).$$

Since k_p has only second-factor covector slots, $k_p(X, X) = k_p(X, Y) = 0$, whereas

$$\begin{aligned} k_p(Y, Y) &= \langle D_p q_t, U q_t \rangle + \langle D_p q'_t, U q'_t \rangle + O(|U|^2) \\ &= \langle D_p, U \rangle_{\text{HS}} + O(|U|^2) = \kappa_p[U] + O(|U|^2). \end{aligned} \tag{5.9}$$

The second equality uses the orthonormal basis $\{q_t, q'_t\}$ of L_0 . The estimates again hold after angular differentiation. Equations (3.9) and (5.7) imply

$$\mathcal{S}_p + \mathcal{L}k_p = O(|U|^2), \quad \mathcal{S}_q + \mathcal{L}k_q = O(|U|^2).$$

This proves the value and differential conditions in (5.5), including derivatives tangent to $\widehat{T}_0$.

Finally, k_p is odd under the first antipodal map and even under the second; k_q has the opposite parities. Both are odd under (5.4). These statements follow directly from (5.6)–(5.8); the radial cutoff preserves them. Hence K_1 is admissible. $\square$

At $p = \pm q$, a graph direction may satisfy $U q_t = 0$ while $U q'_t \neq 0$. The tangent–normal term in (5.9) records precisely this remaining direction. No division by $\sin(t - s)$ occurs in the construction.

From now on, fix any admissible K_1 . Appendix A.3 gives a second, algebraic choice that is useful for direct computation. It need not equal (5.8); only the admissibility conditions will enter the quadratic margin and cubic-mean argument.

5.2 The choice of the perturbation tensors

The corrections involving B, C must not spoil the height-tensor margin. The required compatibility is stronger than equality on the residual torus.

Proposition 5.4 (Vanishing transverse jets). *Each of the following five functions on $\mathcal{Z}_1$ vanishes on $\widehat{T}_0$, together with its first derivative along $\mathcal{Z}_1$:*

$$\begin{aligned} &2\mathcal{Q}(O, B) + \mathcal{L}K_1, \quad 2\mathcal{Q}(O, C), \\ &\mathcal{Q}(B) + \mathcal{L}K_2, \quad 2\mathcal{Q}(B, C) + \mathcal{L}K_3, \quad \mathcal{Q}(C). \end{aligned} \tag{5.10}$$

Proof. The reflection (5.4) fixes $\widehat{T}_0$ pointwise and reverses the four transverse directions in $\mathcal{Z}_1$. Under this reflection O, C, K_0, K_2 are even, whereas B, K_1, K_3 are odd. The exact identities on $\widehat{T}_0$ are

$$\mathcal{Q}(B) = \frac{128}{845} \cos 2\theta, \quad \mathcal{L}K_2 = -\frac{128}{845} \cos 2\theta, \quad \mathcal{Q}(C) = \mathcal{Q}(O, C) = 0. \quad (5.11)$$

Thus the even functions in (5.10) have both zero value and zero first transverse derivative.

The O - B assertion is exactly Definition 5.2. For the remaining odd function, parity gives zero value but not zero derivative. Set

$$D_j = 2\alpha_1 \odot \alpha_2, \quad D_k = 2\alpha_1 \odot \alpha_3, \quad S(U, V) = U g_0^{-1} V + V g_0^{-1} U.$$

Let $A_{\mu\nu}$ send e_μ to e_ν , e_ν to $-e_\mu$, and annihilate the other basis vectors. For any of $A_{02}, A_{03}, A_{12}, A_{13}$, let Φ_a be its diagonal $SO(4)$ -flow and define

$$\dot{B} = \left. \frac{d}{da} \right|_0 \Phi_a^* B, \quad \dot{C} = \left. \frac{d}{da} \right|_0 \Phi_a^* C.$$

Since Φ_a preserves g_0 and $\mathcal{Z}$, the operators $\mathcal{L}, \mathcal{Q}$ are equivariant under this pullback, and the differentiated tensors belong to $\ker \mathcal{L}$. The derivative along the induced flow on $\mathcal{Z}_1$ is therefore

$$\begin{aligned} \left. \frac{d}{da} \right|_0 \Phi_a^* (2\mathcal{Q}(B, C) + \mathcal{L}K_3) &= 2\mathcal{Q}(\dot{B}, C) + 2\mathcal{Q}(B, \dot{C}) \\ &\quad + \frac{8}{5} \mathcal{L}S(\dot{B}, C) + \frac{8}{5} \mathcal{L}S(B, \dot{C}). \end{aligned}$$

For a quaternionic flow $p \mapsto e^{a\ell} p e^{ab}$, the left coframe satisfies $\dot{\alpha} = -[b, \alpha]$. The four rotations above give

$$\begin{array}{c|cccc} & A_{02} & A_{03} & A_{12} & A_{13} \\ \hline b & j/2 & k/2 & -k/2 & j/2 \\ \dot{C} & -D_k & D_j & -D_j & -D_k \end{array}$$

because $\dot{C} = 2\alpha_1 \odot \dot{\alpha}_1$.

We first dispose of the terms containing $\dot{B}$ and C . The reflection (5.4) preserves $g_0 + \tau C$, so T_0 remains totally geodesic for small τ . Its induced metric has constant coefficients $G_1 + \tau \operatorname{diag}(1, 0)$ in s, t . Consequently, for $X = E_i, Y = F_i$ along T_0 ,

$$D_C(X, X) = D_C(X, Y) = D_C(Y, Y) = 0 \quad \text{as full ambient vectors.}$$

Flatness and total geodesicity also give zero differential in the plane directions for every metric in this path. Since $F_{1,C}|_{\mathcal{Z}} = 0$, these directions together with $T\mathcal{Z}$ imply $dF_{1,C} = 0$ on $\widehat{T}_0 \cap \mathcal{Z}_1^{\text{reg}}$. Continuity extends this to all of $\widehat{T}_0$. Formula (3.8) now gives $\mathcal{Q}(\dot{B}, C) = 0$ there.

The tangential components of $\Phi_a^* B$ on T_0 are constant in s, t , since Φ_a maps it to a type-I flat torus. Hence the same is true of $\dot{B}$. Along T_0 , the g_0 -normal space is spanned by E_j, E_k, F_j, F_k , and C annihilates this space. It follows that the tangential restriction of $S(\dot{B}, C)$ has constant coefficients as well. Thus $\mathcal{L}S(\dot{B}, C) = 0$ by (3.9).

The two identities

$$2\mathcal{Q}(B, D_j) = \frac{128}{325} \cos 2\theta, \quad \mathcal{L}S(B, D_j) = -\frac{16}{65} \cos 2\theta, \quad (5.12)$$

$$2\mathcal{Q}(B, D_k) = \frac{128}{325} \sin 2\theta, \quad \mathcal{L}S(B, D_k) = -\frac{16}{65} \sin 2\theta \quad (5.13)$$

cancel the remaining terms for each of the four values of $\dot{C}$, since $128/325 - (8/5)(16/65) = 0$.

The exact B - C verification is given in Appendix A. The four rotations span $\operatorname{Hom}(L_0, L_0^\perp)$. Since $\widehat{T}_0$ is the fiber over L_0 of the supporting-plane map $\mathcal{Z}_1 \rightarrow \operatorname{Gr}_2(\mathbb{R}^4)$, they cover its four transverse directions, including above $p = \pm q$. The function vanishes on $\widehat{T}_0$, so its tangential derivatives vanish too. $\square$

Define the two corrected functions

$$\begin{aligned} F &= 2\mathcal{Q}(O, B) + \mathcal{L}K_1 + 2\mathcal{Q}(O, C), \\ G &= \mathcal{Q}(B) + \mathcal{L}K_2 + 2\mathcal{Q}(B, C) + \mathcal{L}K_3 + \mathcal{Q}(C). \end{aligned}$$

In the graph coordinates $L = L_U$ from Section 5.1, orthogonal projection gives

$$\rho(L_U) = \text{tr}[U^T U (I + U^T U)^{-1}] = \|U\|_{\text{HS}}^2 + O(\|U\|_{\text{HS}}^4).$$

Proposition 5.4 and Taylor's theorem give $|F| + |G| \leq C\|U\|_{\text{HS}}^2$ near $\widehat{T}_0$, uniformly in its compact set of angular variables. Hence $|F| + |G| \leq C'\rho$ there. On the compact complement of this neighborhood, ρ has a positive minimum. Thus $F, G = O(\rho)$ uniformly on $\mathcal{Z}_I$, and

$$C_* = 1 + \sup_{\mathcal{Z}_I \setminus \widehat{T}_0} \frac{|F| + |G|}{\rho} < \infty. \quad (5.14)$$

Choose, in this order,

$$\varepsilon = \frac{1}{1000C_*}, \quad h_* = O + \varepsilon(B + C), \quad K = K_0 + \varepsilon K_1 + \varepsilon^2(K_2 + K_3). \quad (5.15)$$

Then $\varepsilon \leq 1$, and

$$\mathcal{Q}(h_*) + \mathcal{L}K \geq \frac{\rho}{250} - \varepsilon(|F| + |G|) \geq \frac{3}{1000}\rho \quad \text{on } \mathcal{Z}_I. \quad (5.16)$$

Next set

$$M_* = 1 + \|2\mathcal{Q}(H_0, h_*)\|_{\infty, \mathcal{Z}_{II}} + \|\mathcal{Q}(h_*) + \mathcal{L}K\|_{\infty, \mathcal{Z}_{II}}, \quad (5.17)$$

$$c_* = 20M_*, \quad H = h_* + c_* H_0. \quad (5.18)$$

The constants depend only on smooth quantities on the compact background zero locus.

Proposition 5.5. *For H, K in (5.15)–(5.18),*

$$V^{(2)} := \mathcal{Q}(H) + \mathcal{L}K \geq 0, \quad \{V^{(2)} = 0\} = \widehat{T}_0. \quad (5.19)$$

Its first derivative along $\mathcal{Z}$ vanishes on $\widehat{T}_0$.

Proof. By (4.8), adding $c_* H_0$ does not change the quadratic coefficient on $\mathcal{Z}_I$. There (5.16) is positive off $\widehat{T}_0$, and Propositions 5.1 and 5.4 give zero on $\widehat{T}_0$. On the second component, (4.9) gives the lower bound

$$\frac{c_*^2}{10} - c_* M_* - M_* = 20M_*^2 - M_* > 0.$$

Finally, a smooth nonnegative function has zero differential at each point of its zero set. $\square$

6 The third-order curvature coefficient

The second coefficient $V^{(2)}$ now vanishes only on $\widehat{T}_0$. For the chosen tensors H, K , write $V^{(3)} = \mathcal{R}(H, K)$. Following Brendle and Hung [BH26, Proposition 5.5], we determine the average of $\iota_0^* V^{(3)}$ on T_0 by expanding in the tensor amplitudes. The following calculation gives the one coefficient that survives the symmetry argument; it is not a formula for the full pointwise function $V^{(3)}$.

Proposition 6.1. On $\widehat{T}_0$, with $p = e^{si}$, $q = e^{ti}$, and $\theta = t - s$,

$$\mathcal{R}(B + C, K_2 + K_3) = \frac{512}{375} + \frac{9728}{21125} \cos 2\theta. \quad (6.1)$$

The corrected quadratic coefficient and its first derivative along $\mathcal{Z}_1$ vanish on $\widehat{T}_0$.

Proof. The quadratic statement follows from Proposition 5.4. Common left multiplication by e^{ai} preserves g_0, B, C, K_2, K_3 , so it suffices to compute at $p = 1, q = e^{\theta i}$; the answer applies to every common phase. At a point of $\mathcal{Z}_1^{\text{reg}}$ the first minimizing graph displacement, in the order

$$(a_{E_j}, a_{E_k}, a_{F_j}, a_{F_k}, b_{E_j}, b_{E_k}, b_{F_j}, b_{F_k}),$$

is

$$\begin{aligned} \eta_1 &= (a, b), \\ 65a &= (-96, 32 \cot \theta, -64 + 24 \sin^2 \theta, 24 \sin \theta \cos \theta), \\ 65b &= (64 - 24 \sin^2 \theta, 24 \sin \theta \cos \theta, 96, 32 \cot \theta). \end{aligned} \quad (6.2)$$

The eight stationarity equations and the cubic substitution in (3.4) give, exactly,

$$\frac{115712z^4 - 2048z^2 + 115712}{63375(1 + z^2)^2}, \quad \cos \theta = \frac{1 - z^2}{1 + z^2}, \quad \sin \theta = \frac{2z}{1 + z^2}. \quad (6.3)$$

This is (6.1). The calculation is described in Appendix A, including a separate contracted-curvature verification that does not use either companion cubic evaluator.

The apparent poles in (6.2) come from the choice of fixed-base complement. Lemma 3.2 and the smooth displacement below remove them. Equivalently, the full normal calculation at $p = q = 1$ gives zero quadratic coefficient and cubic coefficient $115712/63375$, agreeing with (6.1). Smoothness gives the identity everywhere on $\widehat{T}_0$. $\square$

We describe the displacement without dividing by $\sin \theta$. Put

$$w_\theta = -\frac{32}{65}(\sin \theta \cos \theta j + \cos^2 \theta k), \quad (6.4)$$

$$U_\theta = \frac{1}{65}(-96E_j - (64 - 24 \sin^2 \theta)F_j + 24 \sin \theta \cos \theta F_k), \quad (6.5)$$

$$V_\theta = \frac{1}{65}((64 - 24 \sin^2 \theta)E_j + 24 \sin \theta \cos \theta E_k + 96F_j). \quad (6.6)$$

For the actual first tensor H , define

$$p_\tau = p, \quad q_\tau = q \exp(\tau \varepsilon w_\theta), \quad \sigma_\tau = \text{span}\{E_i + \tau \varepsilon U_\theta, F_i + \tau \varepsilon V_\theta\} \subset T_{(p_\tau, q_\tau)} M. \quad (6.7)$$

All frame fields in the last expression are evaluated at the moved point.

Lemma 6.2. The curve (6.7) has a minimizing first normal displacement along $\widehat{T}_0$. Its cubic curvature coefficient for $g_0 + \tau H + \tau^2 K$ is a smooth periodic function of s, t , equal to the reduced cubic coefficient.

Proof. Hold $p = e^{si}$ fixed and vary the type-I torus axis by $\dot{u} = k$. For $q = p(\cos \theta + \sin \theta u)$, this tangent variation to $\mathcal{Z}_1$ has

$$q^{-1} \dot{q} = \sin^2 \theta j + \sin \theta \cos \theta k, \quad (\dot{X}, \dot{Y}) = (E_k, F_k).$$

Subtracting $32 \cot \theta / 65$ times this tangent vector from (6.2) gives exactly (6.4)–(6.6). The result is smooth at $\theta = 0, \pi$, and the stationarity equations extend there by continuity in the full normal bundle.

Only B contributes to the first normal displacement on $\widehat{T}_0$. For H_0 this follows from (4.8) and its proof; for C it follows from the ambient connection and differential identities proved in Proposition 5.4. For O , the reflection (5.4) keeps T_0 totally geodesic, so Codazzi gives zero first plane-curvature gradient; regular normal complements and continuity give $dF_{1,O} = 0$ on $\widehat{T}_0$. Thus the displacement for H is ε times that for $B + C$. Since $V^{(2)} = dV^{(2)} = 0$ on $\widehat{T}_0$, Lemma 3.2 identifies the cubic coefficient with the reduced one. The explicit formula makes smoothness and periodicity immediate. $\square$

Denote the function in Lemma 6.2 by

$$V_0^{(3)} = \iota_0^* V^{(3)}, \quad V_0^{(3)}(s, t) = V^{(3)}(\iota_0(e^{si}, e^{ti})).$$

We take averages with respect to the constant background area form on T_0 , normalized to total mass one.

Proposition 6.3. *For every admissible K_1 , the choices (5.15)–(5.18) give*

$$\text{mean}_{T_0} V_0^{(3)} = \frac{512}{375} \varepsilon^3 =: m > 0. \quad (6.8)$$

Proof. Introduce real amplitudes $\lambda, \delta, \mu, \gamma$ and set

$$h = \lambda O + \delta B + \mu C + \gamma H_0, \quad (6.9)$$

$$k = \lambda^2 K_0 + \lambda \delta K_1 + \delta^2 K_2 + \delta \mu K_3. \quad (6.10)$$

The corrected quadratic coefficient and its first derivative vanish on $\widehat{T}_0$ for every amplitude choice. This follows term by term from Propositions 4.2, 5.1, and 5.4. Its pullback to T_0 is therefore an intrinsically defined homogeneous cubic polynomial in these amplitudes. The reflection (5.4) changes δ to $-\delta$, so this polynomial is even in δ .

When $\delta = 0$, the reflection preserves the entire metric family and fixes T_0 , which remains totally geodesic. Its intrinsic curvature coefficients of orders zero, one, and two vanish. The reduced cubic equals its intrinsic cubic, since the first normal displacement is zero. More explicitly, write the intrinsic curvature and area form as

$$K_\tau^{\text{int}} = \tau^3 r_{\text{int}} + O(\tau^4), \quad dA_\tau = (1 + \tau a_1 + O(\tau^2)) dA_0.$$

Gauss–Bonnet gives

$$0 = \int_{T_0} K_\tau^{\text{int}} dA_\tau = \tau^3 \int_{T_0} r_{\text{int}} dA_0 + O(\tau^4).$$

Thus the cubic coefficient has zero average with respect to the background area form. Every monomial not involving δ therefore has zero average.

The remaining possibilities are $\delta^2 \lambda, \delta^2 \mu, \delta^2 \gamma$. The simultaneous antipodal map reverses O, K_1 and preserves the other relevant tensors and the torus area, so the coefficient of $\delta^2 \lambda$ has zero mean. To isolate the $\delta^2 \gamma$ coefficient, set $\lambda = 0$. The first-factor antipodal map then reverses H_0 and preserves B, C, K_2, K_3 , so this coefficient also has zero mean.

It remains to determine the coefficient of $\delta^2 \mu$. Set $\lambda = \gamma = 0, \delta = \mu = 1$. Proposition 6.1 gives mean $512/375$, because $\cos 2(t - s)$ has zero mean. For the actual tensors, $\lambda = 1, \delta = \mu = \varepsilon$, and $\gamma = c_*$. This proves (6.8). $\square$

Only the prescribed derivatives and parity of K_1 entered this argument. At fixed tensor amplitudes, replacing K_1 may change the pointwise function $V_0^{(3)}$, but not its average. If the auxiliary constants are reselected for a new K_1 , the value of ε may also change; the identity (6.8) still holds with that value. The Poisson equation uses the corresponding function $V_0^{(3)}$.

7 A third-order conformal correction

We now carry out the conformal Poisson correction from the Brendle–Hung construction [BH26, Section 6], with the normalization appropriate to our torus metric. The inverse of G_I in (2.7) gives

$$\Delta_T = \frac{29}{30}(\partial_s^2 + \partial_t^2) + \frac{38}{30}\partial_s\partial_t. \quad (7.1)$$

Our Laplacian convention has nonpositive eigenvalues on nonconstant Fourier characters.

By Proposition 6.3, there is a unique smooth mean-zero solution of

$$\Delta_T f = 2(V_0^{(3)} - m). \quad (7.2)$$

For completeness, if $\widehat{V_0^{(3)}}_{ab}$ are the Fourier coefficients in the 2π -periodic coordinates s, t , then

$$f(s, t) = -60 \sum_{(a,b) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{\widehat{V_0^{(3)}}_{ab} e^{i(as+bt)}}{29(a^2 + b^2) + 38ab}. \quad (7.3)$$

The denominator equals $10(a^2 + b^2) + 19(a + b)^2$ and is positive away from the zero character. Smoothness of $V_0^{(3)}$ implies convergence with all derivatives. The usual conjugacy of the Fourier coefficients makes f real-valued.

Here is an explicit smooth extension. Let $a_p = p_0^2 + p_1^2$, $a_q = q_0^2 + q_1^2$, and choose a smooth cutoff χ that is zero on $(-\infty, 1/2]$ and one on $[3/4, \infty)$. Where the denominators are nonzero, set

$$z_p = \frac{p_0 + ip_1}{\sqrt{a_p}}, \quad z_q = \frac{q_0 + iq_1}{\sqrt{a_q}}.$$

Regard f as a function on $S^1 \times S^1$, and define

$$\tilde{f}(p, q) = \chi(a_p)\chi(a_q)f(z_p, z_q), \quad (7.4)$$

setting it equal to zero wherever a projection is undefined. The cutoff vanishes in a neighborhood of each such point, so the extension is smooth on M and equals f on T_0 .

Theorem 7.1 (Perturbation family). *Fix g_0 by (2.1) and the tensors $H_0, O, B, C, K_0, K_2, K_3$ by (4.1)–(4.7). Choose any admissible K_1 from Definition 5.2; such a tensor is supplied by (5.8), or by the algebraic formula (A.8). For this choice, fix the constants and function by (5.14)–(5.18) and (7.2)–(7.4). There is $\tau_0 > 0$ such that*

$$g_\tau = g_0 + \tau H + \tau^2 K + \tau^3 \tilde{f} g_0 \quad (7.5)$$

is a smooth Riemannian metric and

$$\sec_{g_\tau}(\sigma) > 0 \quad \text{for every } x \in S^3 \times S^3, \sigma \in \text{Gr}_2(T_x M), \quad 0 < \tau < \tau_0.$$

Proof. All tensors and constants have been fixed before choosing τ . Since $g_0 \geq (5/8)I_6$ and the perturbation tensors are bounded on the compact manifold, g_τ is positive definite for sufficiently small $|\tau|$.

On T_0 , which is flat and totally geodesic for the background metric, the conformal first-variation formula is

$$\iota_0^* \mathcal{L}(\tilde{f} g_0) = -\frac{1}{2} \Delta_T f.$$

The term of order τ^3 in (7.5) therefore changes the pullback of the reduced cubic to

$$\iota_0^* \tilde{V}^{(3)} = V_0^{(3)} - \frac{1}{2} \Delta_T f = m > 0,$$

where we write $\tilde{V}^{(3)} = V^{(3)} + \mathcal{L}(\tilde{f}g_0)$ on $\mathcal{Z}$. It does not change the second coefficient $V^{(2)}$. The full normal minimum has a uniform expansion

$$\tau^2 V^{(2)}(z) + \tau^3 \tilde{V}^{(3)}(z) + O(\tau^4), \quad z \in \mathcal{Z}, \quad (7.6)$$

where $V^{(2)} \geq 0$, $\{V^{(2)} = 0\} = \hat{T}_0$, and $\tilde{V}^{(3)} = m > 0$ on $\hat{T}_0$. These are precisely the hypotheses of Lemma 3.1, using the positive full normal Hessian in Proposition 2.2. It follows that every plane in a fixed tube about $\mathcal{Z}$ has positive curvature for all sufficiently small positive τ , uniformly in its base point. The compact complement has a positive background curvature minimum, which also persists. Decreasing τ_0 accommodates both regions and proves the asserted strict inequality on the whole Grassmann bundle. $\square$

Theorem 1.1 follows. Since each three-sphere has Euler characteristic zero and Euler characteristic is multiplicative for products of finite CW complexes, this also gives $\chi(S^3 \times S^3) = 0$ for a closed even-dimensional positively curved manifold.

A Exact curvature identities and their verification

This appendix specifies the finite calculations used in Propositions 4.2, 5.1, 5.4, and 6.1. The accompanying code evaluates these identities over $\mathbb{Q}(z)$, not by numerical sampling. The geometric arguments establishing the zero locus, the appropriate normal spaces, and uniform positivity are separate from these algebraic checks.

A.1 Frame calculus

Order the frame as $(e_1, \dots, e_6) = (E_i, E_j, E_k, F_i, F_j, F_k)$, and write $[e_i, e_j] = c_{ij}^m e_m$, summing repeated indices. Write R^g for the curvature endomorphism, to distinguish it from $R = \text{Ad}_{p^{-1}q}$. We use (3.1), so $\text{sec}_g(X \wedge Y) = g(R^g(X, Y)Y, X) / |X \wedge Y|_g^2$. For any metric matrix g , the noncoordinate formulas are

$$2g_{k\ell}\Gamma_{ij}^\ell = e_i g_{jk} + e_j g_{ik} - e_k g_{ij} + c_{ij}^m g_{mk} - c_{jk}^m g_{mi} + c_{ki}^m g_{mj}, \quad (\text{A.1})$$

$$(R^g)_{ijk}^\ell = e_i \Gamma_{jk}^\ell - e_j \Gamma_{ik}^\ell + \Gamma_{jk}^m \Gamma_{im}^\ell - \Gamma_{ik}^m \Gamma_{jm}^\ell - c_{ij}^m \Gamma_{mk}^\ell. \quad (\text{A.2})$$

All spatial two-jets follow from quaternion multiplication and

$$E_a r = -ar, \quad F_a r = ra. \quad (\text{A.3})$$

For example, if $J_a b = a \times b$, then

$$E_a R = -2J_a R, \quad F_a R = 2R J_a, \quad E_b E_a R = 4J_a J_b R.$$

The last order of multiplication is intentional. The code uses ordered frame derivatives and checks their commutators against the structure constants.

For a metric series, the coefficients of the lower connection in (A.1) depend linearly on the metric coefficients. The upper connection coefficients are then obtained recursively by solving against the fixed zeroth-order metric. Differentiating this recurrence and substituting into (A.2) gives the curvature coefficients. This supplies one direct method of computing both the quadratic and cubic identities.

A separate check contracts curvature before expanding the full tensor. For frame fields with constant coefficients, let $B_g(X, Y, Z) = g(\nabla_X Y, Z)$ and $c = [X, Y]$. Metric compatibility

and torsion freeness give

$$\begin{aligned}
g(R^g(X, Y)Y, X) &= X(B_g(Y, Y, X)) - Y(B_g(X, Y, X)) \\
&\quad + B_g(X, Y, \cdot)^T g^{-1} B_g(X, Y, \cdot) \\
&\quad - B_g(X, X, \cdot)^T g^{-1} B_g(Y, Y, \cdot) \\
&\quad - B_g(X, Y, c) - B_g(c, Y, X).
\end{aligned} \tag{A.4}$$

At a background flat torus, the three lower connection vectors in the Gram terms vanish at order zero. Their contribution through order three therefore requires only the first derivative of g^{-1} . The separate rational-function verification uses (A.4); it does not invoke either of the other cubic evaluators.

A.2 The quadratic coefficient of the odd height tensor

At $p = 1$, $q = \cos \theta + i \sin \theta$, write A_p, A_q for the constant matrices multiplying p_0, q_1 in (4.2), and put $X_0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$. For arbitrary $e, f \in \mathbb{R}^4$, substitute

$$\begin{aligned}
h(e, f) &= \langle e, p \rangle A_p + \langle f, q \rangle A_q, \\
k(e, f) &= -\frac{1}{5} \langle e, p \rangle \langle f, q \rangle X_0 - \frac{787}{5880} \langle f, p \rangle \langle e, q \rangle I_6.
\end{aligned}$$

The result is a quadratic expression

$$\mathcal{Q}(h(e, f)) + \mathcal{L}k(e, f) = e^T \mathbf{M}_p e + f^T \mathbf{M}_q f + e^T \mathbf{M}_{pq} f.$$

With the polynomials in Proposition 5.1, exact substitution gives

$$\begin{aligned}
\mathbf{M}_p = \mathbf{M}_q &= \text{diag} \left(0, 0, \frac{P}{2d}, \frac{P}{2d} \right), \\
\mathbf{M}_{pq} &= d_0 I_4 + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & V \cos \theta / d & -W \sin \theta / d \\ 0 & 0 & W \sin \theta / d & V \cos \theta / d \end{pmatrix}, \quad d_0 = -\frac{787}{11025} \cos \theta.
\end{aligned} \tag{A.5}$$

The verification checks every entry of these three matrices as a rational function. The orthogonality $e \perp f$, supplied by the diagonal $SO(4)$ -normalization, eliminates the $d_0 I_4$ term. This proves the identity for every type-I torus, not just one choice of its ambient position.

A.3 An algebraic mixed correction

The proof in Section 5.1 does not require a closed algebraic formula for K_1 . We record a second admissible choice for the executable certificate. It is independent of the integral construction, not asserted to be the same tensor.

Let $\ell, \widehat{\ell}$ be either coframe pair $(\alpha, \widehat{\alpha})$ or $(\beta, \widehat{\beta})$, and put

$$\begin{aligned}
U_\ell &= \ell_1 + \widehat{\ell}_1, & V_\ell^\pm &= \ell_2 \pm \widehat{\ell}_2, & W_\ell^\pm &= \frac{13}{7} \ell_3 \pm \widehat{\ell}_3, \\
T_\ell &= \ell_1^2 + \ell_1 \odot \widehat{\ell}_1 + \widehat{\ell}_1^2.
\end{aligned}$$

For a quaternion coordinate vector $x = (x_0, x_1, x_2, x_3)$, define

$$\mathcal{A}_\ell(x) = U_\ell \odot (x_0 V_\ell^- + x_1 W_\ell^-) - \frac{4}{7} x_3 T_\ell, \tag{A.6}$$

$$\mathcal{B}_\ell(x) = U_\ell \odot (x_0 W_\ell^+ - x_1 V_\ell^+) - \frac{4}{7} x_2 T_\ell. \tag{A.7}$$

Set

$$K_1^{\text{alg}} = \frac{11}{325}(\mathcal{A}_\beta(p) + \mathcal{B}_\alpha(q)) + \frac{6}{65}r_0(\mathcal{B}_\alpha(p) + \mathcal{A}_\beta(q)). \quad (\text{A.8})$$

In $\mathcal{A}_\ell(x)$ and $\mathcal{B}_\ell(x)$, x supplies the scalar coordinate functions, while ℓ specifies the coframe in the tensor factors. Direct substitution shows that this tensor is odd under (5.4) and the simultaneous antipodal map. We verify the remaining first-jet condition below.

The B - C identities (5.12)–(5.13) are direct evaluations of (3.8) and (3.9). Here we explain the global coverage of the O - B check.

Write $p = e^{\varphi i}$, $q = e^{(\varphi+\theta)i}$, and define

$$\begin{aligned} a &= -\frac{64}{455} \sin \theta \cos \theta, & b &= \frac{32}{6825} (41 - 30 \sin^2 \theta), \\ d &= \frac{352}{6825} \sin \theta, & e &= \frac{1312}{6825} \cos \theta. \end{aligned}$$

Let A_{ab} denote the infinitesimal rotation taking e_a toward e_b . In the order $A_{02}, A_{03}, A_{12}, A_{13}$, the first transverse derivatives of $2\mathcal{Q}(O, B)$ are

$$(a \cos \varphi - b \sin \varphi, \quad b \cos \varphi + a \sin \varphi, \quad d \cos \varphi + e \sin \varphi, \quad -e \cos \varphi + d \sin \varphi). \quad (\text{A.9})$$

Restriction of (A.8) and application of (3.9) give their negatives.

For a type-I torus, let u, w denote its constant left and right great-circle axes. At T_0 , both equal i . The restrictions of the local expressions (A.6)–(A.7) have transverse derivatives

$$\dot{\mathcal{A}}_\ell(x)(T, T) = 2\{x_0(\dot{u}_j - \dot{w}_j) + x_1(13\dot{u}_k/7 - \dot{w}_k)\} - \frac{12}{7}\dot{x}_3, \quad (\text{A.10})$$

$$\dot{\mathcal{B}}_\ell(x)(T, T) = 2\{x_0(13\dot{u}_k/7 + \dot{w}_k) - x_1(\dot{u}_j + \dot{w}_j)\} - \frac{12}{7}\dot{x}_2. \quad (\text{A.11})$$

Here T is the round-unit-speed great-circle tangent on the factor carrying ℓ , not a vector rescaled to have unit g_0 -length. For $A_{02}, A_{03}, A_{12}, A_{13}$, respectively, the pairs $(\dot{u}, \dot{w})$ are

$$(k, -k), \quad (-j, j), \quad (j, j), \quad (k, k).$$

They are independent of the common phase. Indeed, a fixed infinitesimal $SO(4)$ -action has the form $\dot{p} = ap + pb$; differentiation of $u = p^{-1}\dot{p}_{\text{circle}}$ and $w = \dot{p}_{\text{circle}}p^{-1}$ gives constant commutator variations of the axes.

The scalar coefficients in K_1^{alg} restrict either to a coordinate on the opposite factor or to r_0 times a coordinate on the same factor. In each case, the second derivative appearing in (3.9) equals the negative of that coefficient. Thus restriction and linearization contribute the factor $1/(2D_1)$ to the expressions (A.10)–(A.11). Substituting the four axis pairs gives the negatives of (A.9).

When differentiating $\mathcal{Q}(O, B)$ under the diagonal action, the term $\mathcal{Q}(O, \dot{B})$ vanishes on $\hat{T}_0$. The reflection (5.4) keeps T_0 totally geodesic for $g_0 + \tau O$, so $D_O(X, Y)$ is tangent to T_0 for $X, Y \in TT_0$. As in Lemma 6.2, $dF_{1,O} = 0$ on $\hat{T}_0$. The tangential components of $\dot{B}$ in the round circle coordinates are constant, whence

$$(D_{\dot{B}}(X, Y))^{\top} = 0, \quad X, Y \in TT_0.$$

This assertion concerns the tangential projection, not the full vector. The connection inner products and Schur term in (3.8) therefore all vanish. This justifies computing the derivative through the height functions alone.

The verification treats θ symbolically throughout and uses the four exact common phases

$$(\cos \varphi, \sin \varphi) \in \{(1, 0), (0, 1), (3/5, 4/5), (4/5, 3/5)\}.$$

The axis variations are independent of the common phase. Moreover, g_0 and B are invariant under common left multiplication by $e^{i\varphi}$, while the height functions transform linearly in $\cos \varphi, \sin \varphi$. Thus the height-derivative expressions are linear in these two functions. The four evaluations are also injective on the larger space $\text{span}\{\cos \varphi, \sin \varphi, \cos 3\varphi, \sin 3\varphi\}$. Indeed, for a polynomial with coefficients a_1, b_1, a_3, b_3 in this order, the first two zero evaluations give $a_1 = -a_3, b_1 = b_3$. The remaining equations have coefficient matrix

$$\frac{1}{125} \begin{pmatrix} -192 & 144 \\ -144 & 192 \end{pmatrix},$$

which is nonsingular. Hence all four coefficients vanish. The four rotations span the required transverse directions, and reflection oddness gives zero value on $\widehat{T}_0$. Thus K_1^{alg} satisfies Definition 5.2.

A.4 Normal complements at the exceptional strata

At the type-II representative $p = 1, q = j, u = i$, use the base curve

$$r(z) = \left(\frac{2z}{1+z^2}, 0, \frac{1-z^2}{1+z^2}, 0 \right)$$

together with the eight graph variables of (3.8). The resulting nine-variable Hessian H_{II} has kernel

$$\ker H_{\text{II}} = \text{span}\{\partial_{a_{E_k}} + \partial_{b_{F_k}}\}.$$

This is the common-axis variation of u toward k , with p, q fixed. The rank calculation in Section 3 shows that there are no other kernel directions in this nine-dimensional slice. Setting $b_{F_k} = 0$ is transverse to this kernel and selects an eight-dimensional normal complement. Let A be the Hessian restricted to that complement and r_{H_0} the corresponding differential of the first unnormalized curvature coefficient. The full nine stationarity equations are checked, and the Schur term equals

$$\frac{1}{2} r_{H_0}^T A^{-1} r_{H_0} = \frac{5107}{151875}.$$

The projection of Z_{II} to M is the hypersurface $\langle p, q \rangle = 0$. The z -direction supplies the one normal direction missing from the fixed-base calculation.

At $p = q = 1$, use two base parameters

$$q(z_j, z_k) = \frac{1 - z_j^2 - z_k^2 + 2z_j j + 2z_k k}{1 + z_j^2 + z_k^2}.$$

Together with the eight graph variables these give ten variables. Their Hessian H_{Δ} has kernel

$$\ker H_{\Delta} = \text{span}\{\partial_{a_{E_j}} + \partial_{b_{F_j}}, \partial_{a_{E_k}} + \partial_{b_{F_k}}\}.$$

These are the two common-axis variations with the base point fixed. They describe the kernel within this ten-dimensional slice, not all of TZ_{I} . The conditions $b_{F_j} = b_{F_k} = 0$ intersect the kernel trivially and therefore give an eight-dimensional normal complement. In the order $(a_{E_j}, a_{E_k}, a_{F_j}, a_{F_k}, b_{E_j}, b_{E_k}, b_{F_j}, b_{F_k}, z_j, z_k)$, the first-order minimizing displacement for $B + C$ is

$$\frac{1}{65}(-192, 0, -64, 0, 64, 0, 0, 0, 0, -16). \quad (\text{A.12})$$

It satisfies all ten stationarity equations. Substitution of the metric with correction $K_2 + K_3$ gives quadratic coefficient zero and cubic coefficient 115712/63375. The mixed quadratic

term between the two base parameters is zero by the circle isotropy rotating the j, k -plane of the background.

The same coefficient is obtained along the smooth curve

$$q(\tau) = \frac{1 - a^2\tau^2}{1 + a^2\tau^2} - \frac{2a\tau}{1 + a^2\tau^2}k, \quad a = \frac{16}{65},$$

and

$$X_\tau = E_i + \frac{\tau}{65}(-96E_j - 64F_j), \quad Y_\tau = F_i + \frac{\tau}{65}(64E_j + 96F_j).$$

Composing the full spatial metric two-jets with this base curve before applying the curvature recurrence gives the exact coefficients $(0, 0, 0, 115712/63375)$. The graph-stationarity derivatives vanish as well. This is the smooth representative from Lemma 6.2, which extends across the diagonal and antidiagonal.

A.5 The generic cubic and the accompanying computations

For the generic cubic calculation, substitute (6.2) into (3.4), with $h = B + C$, $k = K_2 + K_3$. The zeroth, first, and second coefficients vanish exactly, and the third is (6.3). A separate use of (A.4) gives the same rational function and checks the eight first-displacement stationarity equations. Because the lower three coefficients vanish, division by the varying Gram determinant does not change the cubic coefficient except for division by D_1 . This qualification is necessary; it would not be valid at a point with a nonzero reduced second coefficient.

The [versioned companion repository](#), at tag `v0.1.0-paper-2026-09-05`, contains the two-jet definitions, curvature evaluators, assertions, accepted outputs, and exact manuscript PDF. After installing its pinned requirements, run `python3 verification/verify.py` from the repository root. Success means exit status zero and a final line beginning `VERIFIED: CLAIM-MAP.md` associates each computer-assisted assertion with its paper location, script, output, and limitations; the repository manifest fixes the artifact identities. The primary verifier checks the first family, the second family, and the full normal calculation at the diagonal; supplemental files check the compact K_1^{alg} formula and the second-order coefficient of the height tensor O there. Further files contain the separate contracted calculation and the smooth diagonal calculation.

The integral correction (5.8) is justified by Proposition 5.3, not evaluated by the algebraic verifier. The main theorem applies to either admissible implementation; its auxiliary bounds and final Poisson right-hand side are evaluated for the implementation chosen.

These programs verify finite algebraic identities; they are not a proof-assistant formalization of the geometric argument. The contracted calculation shares the primary metric and tensor two-jets, while a separately differentiated coordinate implementation gives additional floating-point checks. Those numerical checks are corroborative only. The logical use of the algebraic certificates is the finite set of identities stated above; the deductions from them to positivity on all two-planes are the geometric proofs in Sections 2–7.

A.6 Choice of coefficients and auxiliary parameters

There are two kinds of choices in the construction. The background parameters and the rational coefficients in the tensor formulas specify one explicit construction. They must satisfy the vanishing and sign conditions proved above; no optimality of these coefficients is asserted. Once these tensors are fixed, the remaining scale parameters are chosen from strict inequalities, in the order given in Section 5. They are not numerical outputs of the verification programs.

Some of the rational coefficients are determined directly by cancellation. For example, after fixing B , consider the correction $a(2r_0^2 - 1)g_0$. On T_0 , we have $2r_0^2 - 1 = \cos 2(t - s)$, and (7.1) gives

$$\Delta_T \cos 2(t - s) = -\frac{8}{3} \cos 2(t - s).$$

Consequently its linear curvature contribution, pulled back to T_0 , is $\frac{4}{3}a \cos 2(t - s)$. Cancelling the coefficient $\frac{128}{845} \cos 2(t - s)$ in (5.11) requires

$$\frac{128}{845} + \frac{4}{3}a = 0, \quad a = -\frac{96}{845},$$

which explains the coefficient of K_2 . Similarly, the two transverse identities (5.12)–(5.13) impose the same scalar equation

$$\frac{128}{325} - \frac{16}{65}b = 0, \quad b = \frac{8}{5},$$

for the coefficient of $Bg_0^{-1}C + Cg_0^{-1}B$ in K_3 . The integral construction of K_1 instead solves the linear equations (5.7): the integral operator in (5.6) gives an explicit solution.

The auxiliary constants are deliberately non-sharp. The choice $\varepsilon C_* = 1/1000$ leaves the lower bound $(1/250 - 1/1000)\rho = 3\rho/1000$ on the first component. The choice $c_* = 20M_*$ makes $c_*^2/10 - c_*M_* - M_* > 0$ on the second component. Finally, τ_0 is obtained from the uniform estimates on compact sets, after all other data have been fixed. No numerical value or maximality property of τ_0 is used.

The displayed cancellation equations explain the correction coefficients. The exploratory selection of the initial background and the remaining rational coefficients is not reconstructed here. These fixed data are the inputs to the exact verifications described above.

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